



Aerospace Environment

Module 2: The Atmosphere

Xinlin Li

[Comments on the Quizzes \(or mini-exams\)](#)

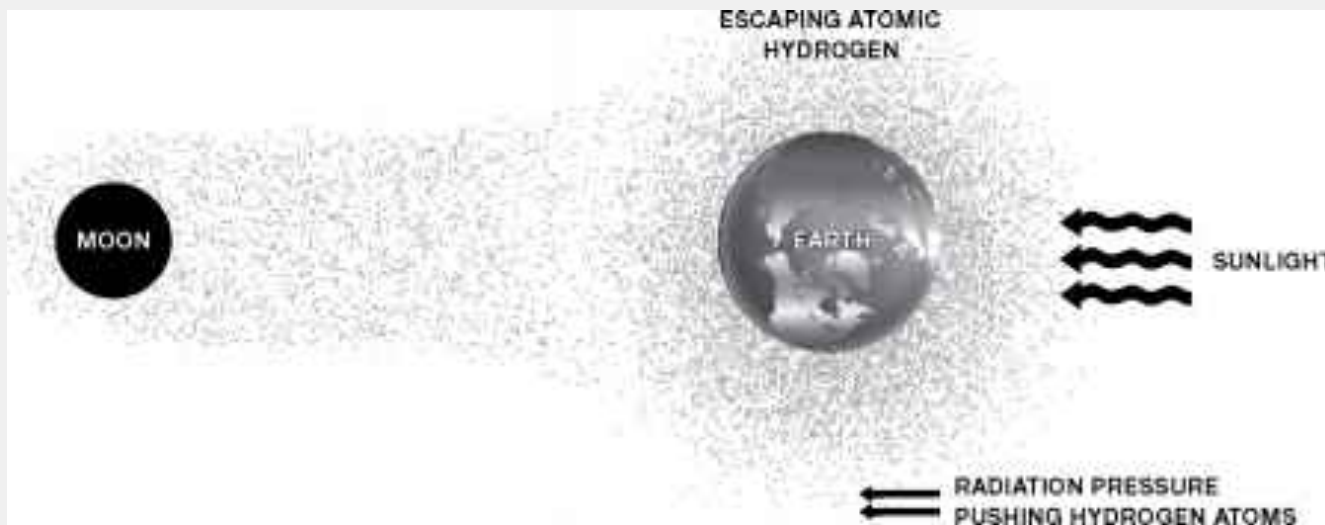


Goals of This Module

- ❖ **Understand basic physics of the Earth's atmosphere**
 - ❖ Atmosphere composition, layers, regions, density
 - ❖ Atmospheric dynamics ([variations](#)), winds, tides, and waves
 - ❖ Atmospheric response to geomagnetic activity
- ❖ **Understand effects of the Atmosphere on Spacecraft operations**
 - ❖ Satellite drag
 - ❖ Oxygen effects
 - ❖ Other effects: erosion and sputtering
- ❖ **Learn about Atmospheric space missions, including CHAMP and GRACE**

Geocorona

- ❖ Earth's atmosphere is visible out to at least 15 Earth radii ($15 R_E$)
- ❖ Scattering of solar Lyman- α (121.4 nm) from hydrogen in the exosphere
- ❖ exosphere theoretically extends to $30 R_E$, where solar radiation pressure exceeds Earth's gravity
- ❖ where is the outer boundary of Earth's atmosphere (where the pressure of SRP 'balances' Earth's gravity)?

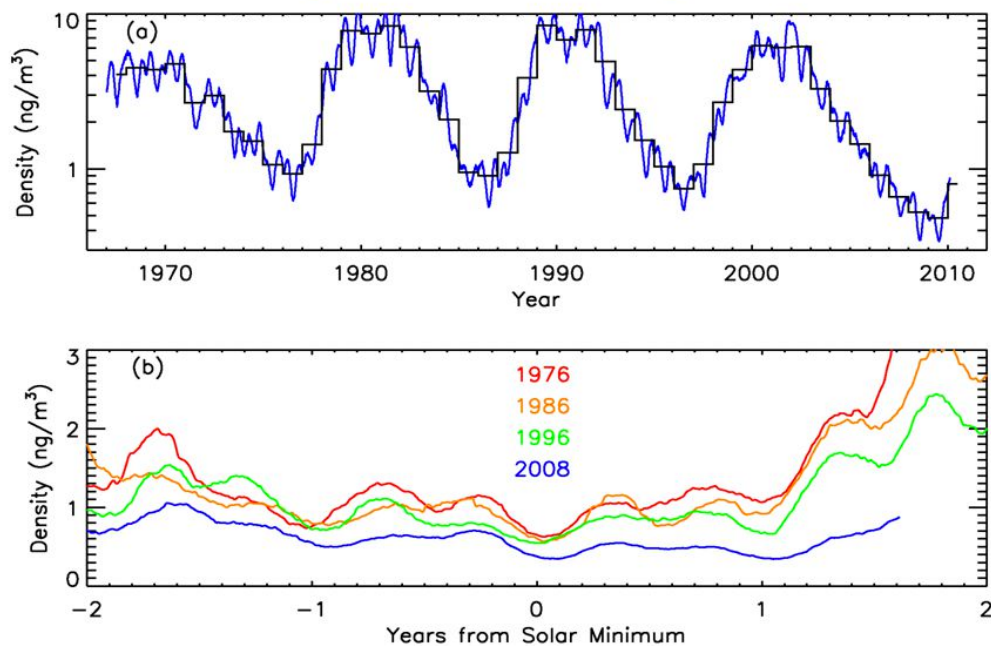


Atmospheric Variability

Solar-Cycle Variation

- ❖ Increase in solar EUV increases heating in the atmosphere, especially thermosphere;
- ❖ Increased heat => increased ionization (next module!)
- ❖ Increase in temperature => atmosphere rises, higher thermosphere density at a given altitude

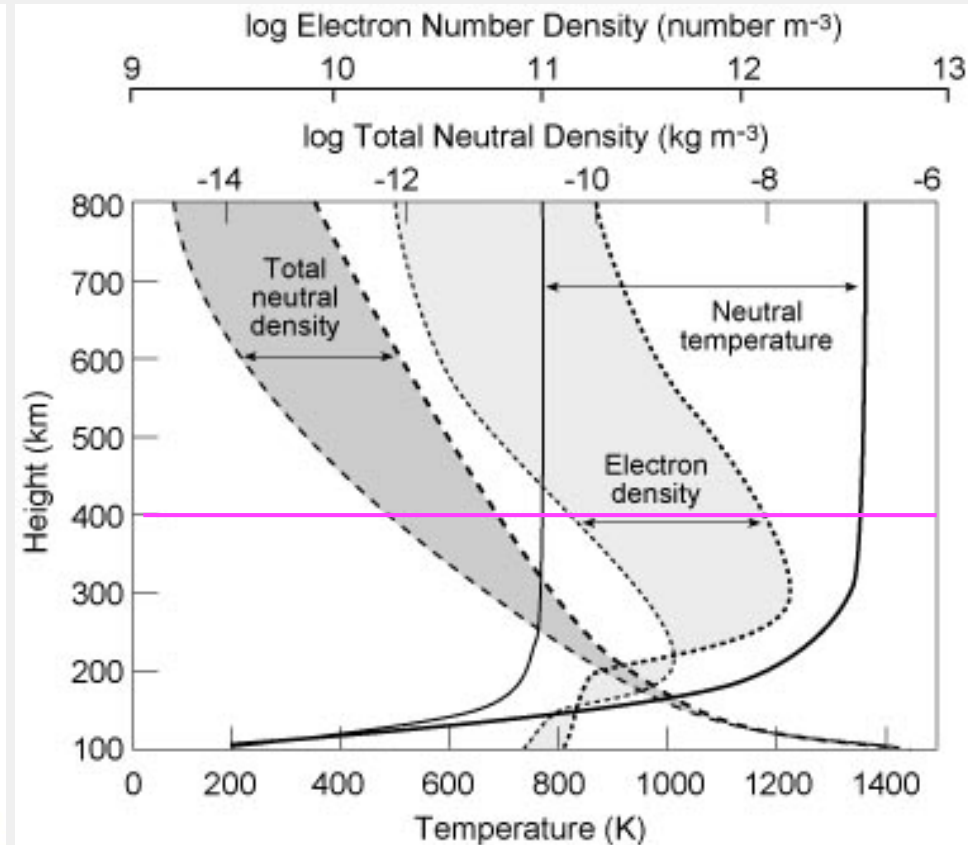
Long-Term Satellite Drag Data



Global average neutral density at 400 km, 81-day average and annual average (top), and superposed epoch analysis (bottom)

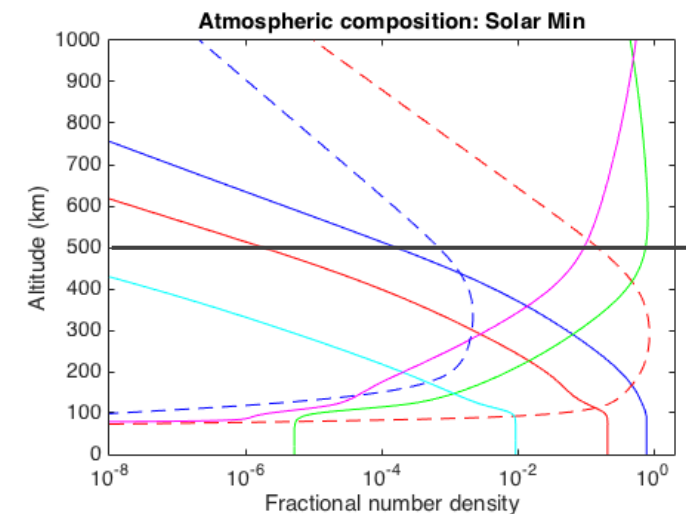
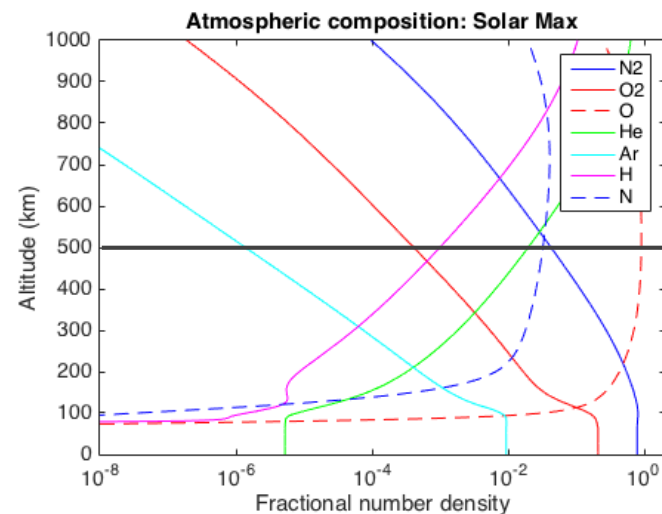
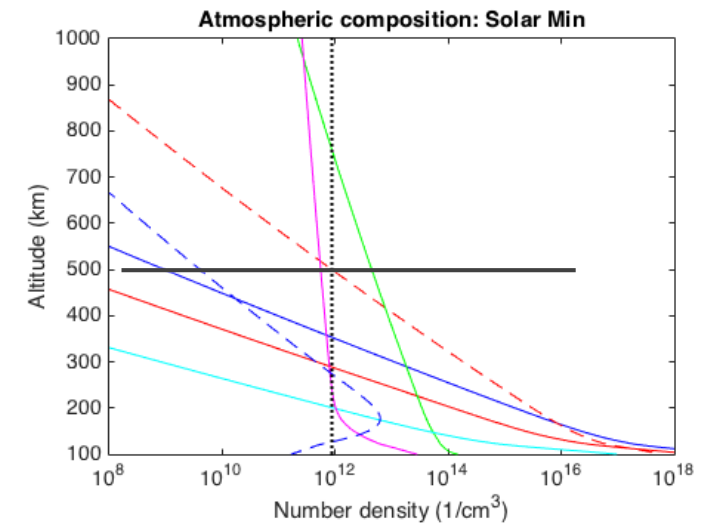
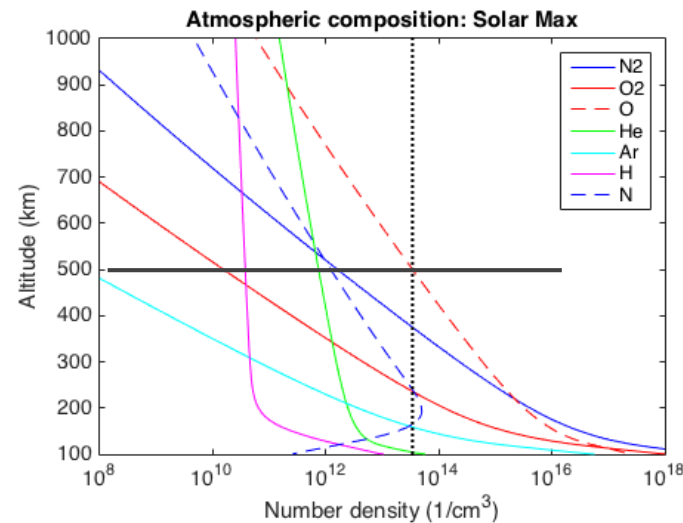
Emmert et al., Geophys. Res. Lett., 37, L12102, 2010

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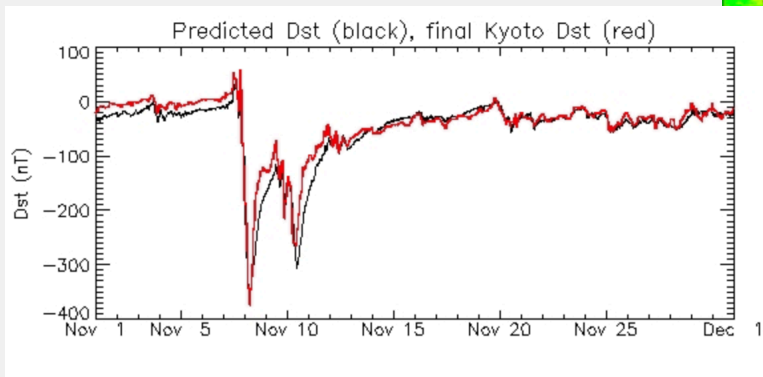
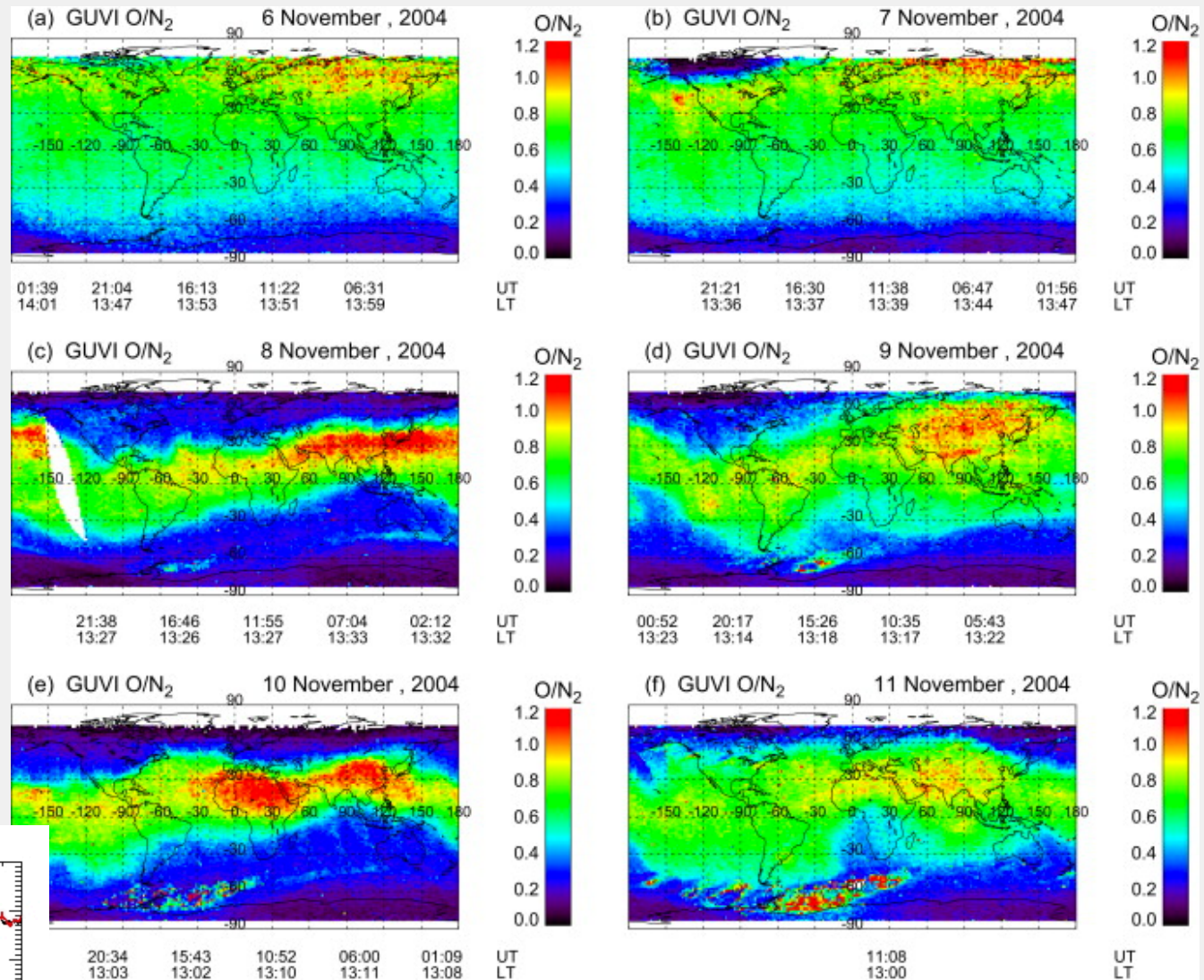
Composition Variation

- ❖ Change in temperature in the thermosphere (from 500 to 1500 K) affects diffusive equilibrium of components
- ❖ Generally, all particles get pushed to higher altitudes by higher temperatures below



GUVI (Global UV Imager) O/N2 ratio

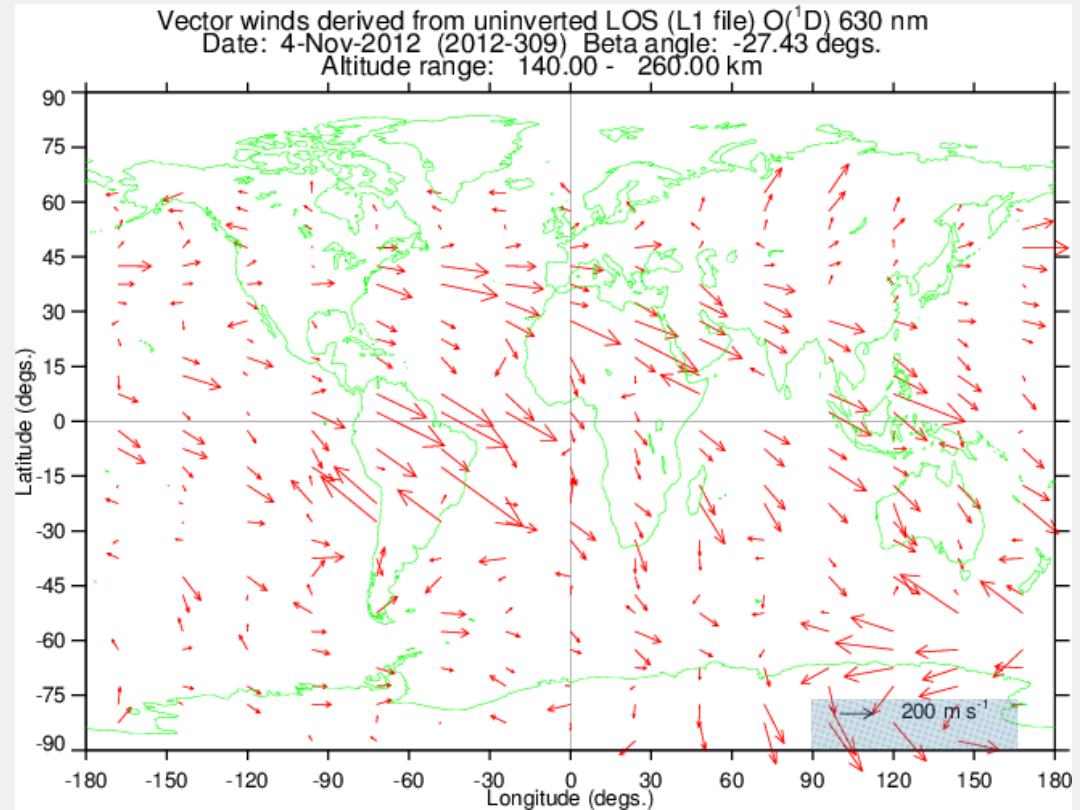
- ❖ GUVI (on TIMED satellite) measured the column density ratio of O to N₂; used as a proxy for neutral composition / heating of the atmosphere



Winds

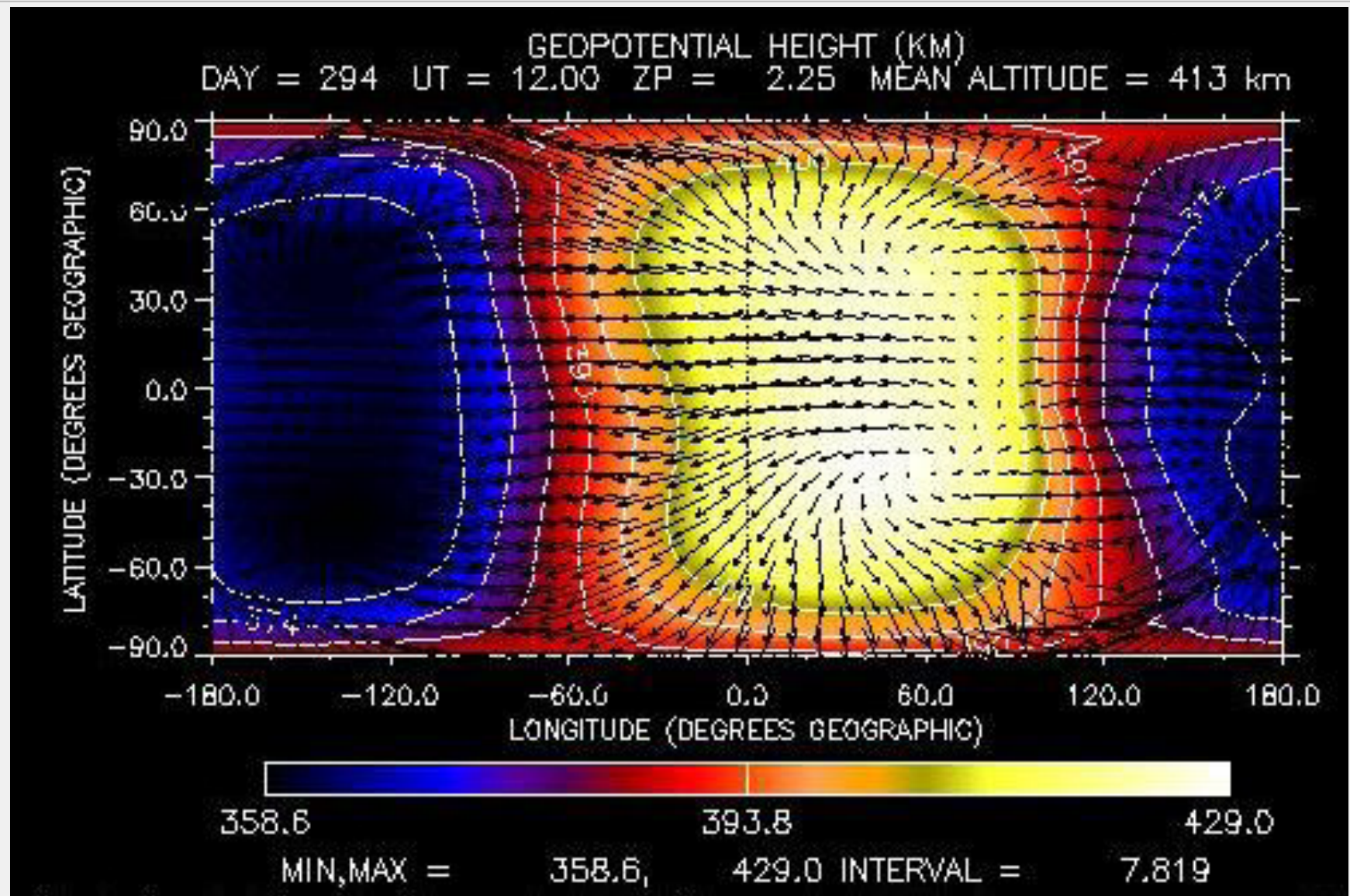
- ❖ Winds in the upper atmosphere are driven by at least four factors:
 - ❖ Pressure gradient, day-to-night
 - ❖ Coriolis force
 - ❖ Viscosity
 - ❖ Ion drag
- ❖ Different effects dominate in different regions of atmosphere (altitude, latitude)
- ❖ In Math:

$$\frac{D\mathbf{U}_h}{Dt} = -\frac{1}{\rho} \nabla_h p - 2\boldsymbol{\Omega} \times \mathbf{U}_h + \frac{1}{\rho} \nabla(\mu \nabla \mathbf{U}_h) - \nu_{ni}(\mathbf{U}_h - \mathbf{V}_i)$$



from TIDI (TIMED Doppler Interferometer)
instrument on TIMED spacecraft

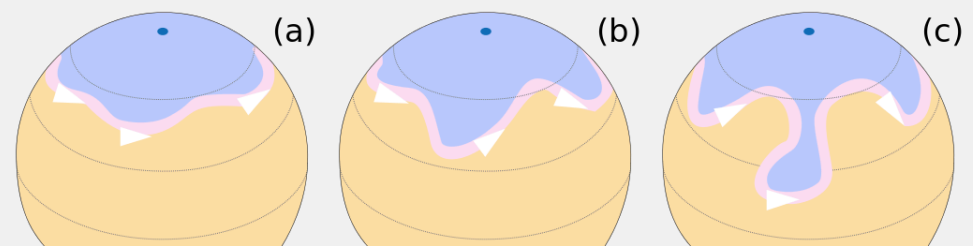
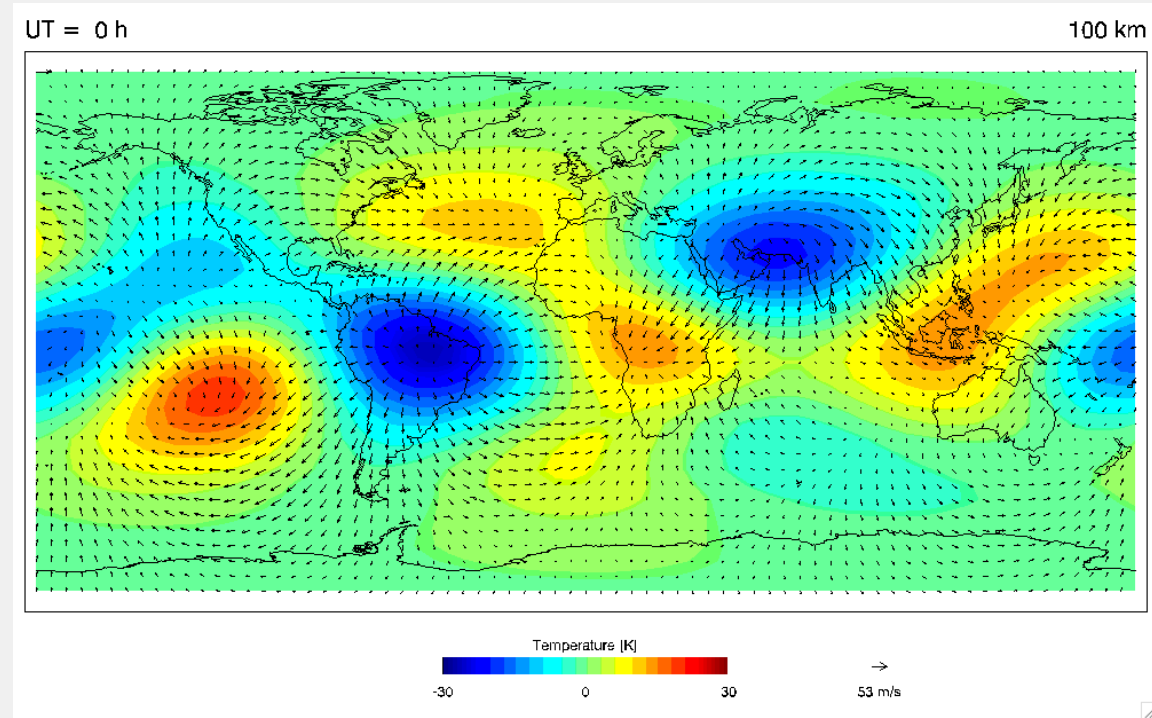
More looks at Thermospheric winds



pressure gradient dominated;
wind flows day to night

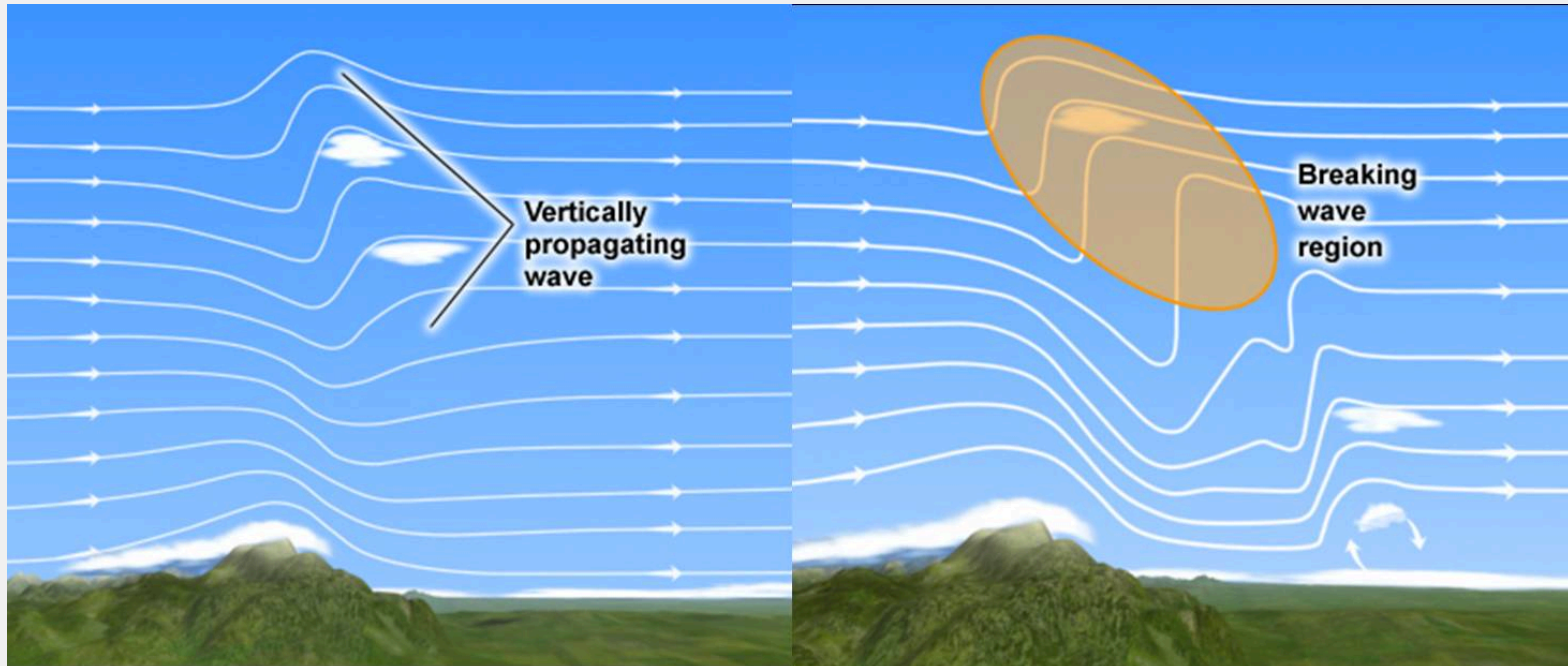
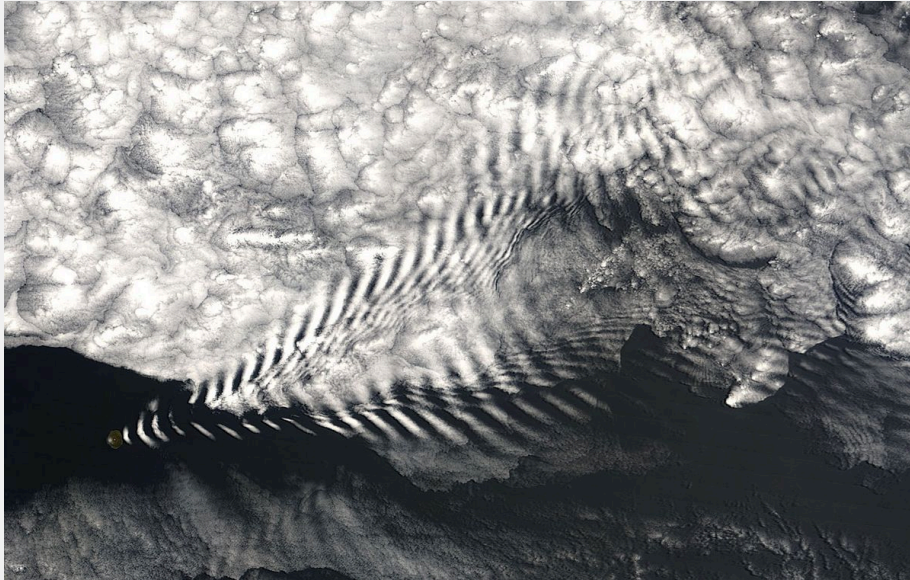
Tides

- ❖ Periodic oscillations in the atmosphere; just like ocean tides (but higher)
- ❖ **Caused by:**
 - ❖ Regular day-night cycle in the Sun's heating of the atmosphere
 - ❖ The gravitational pull of the Moon
 - ❖ Non-linear interactions between tides and *planetary waves*
 - ❖ Large-scale latent heat release due to deep convection in the tropics (thunderstorms)
- ❖ Can be migrating (sun-synchronous) or non-migrating
- ❖ Because atmosphere density falls with altitude, tide amplitude increases with altitude – exponentially!



Planetary (or Rossby) waves

Gravity Waves



Gravity Waves in upper atmosphere

- ❖ Similar to acoustic waves; derived from fundamental fluid equations
- ❖ Difference: **gravity (g) matters**. Allows waves to propagate with vertical direction
 - ❖ Bouyancy waves
- ❖ Amplitude is inversely proportional to pressure: grows exponentially with altitude
- ❖ Observed as temperature and/or density oscillations at particular altitudes

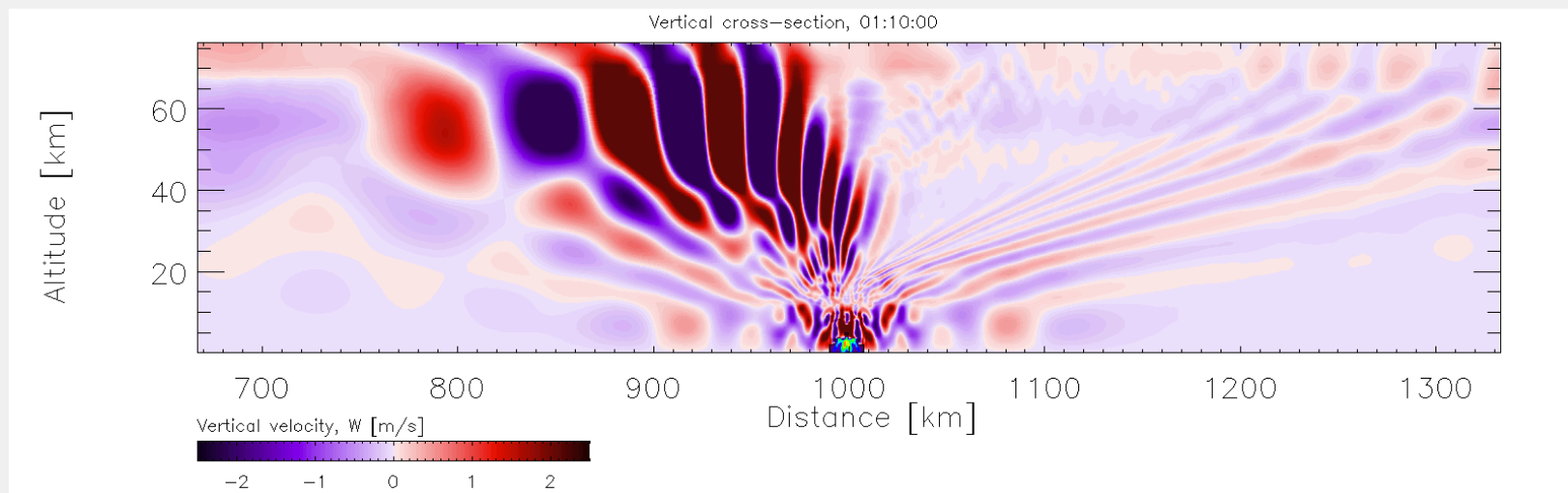
$$\frac{du}{dt} - fv + \frac{1}{\rho} \frac{\partial p}{\partial x} = X,$$

$$\frac{dv}{dt} + fu + \frac{1}{\rho} \frac{\partial p}{\partial y} = Y,$$

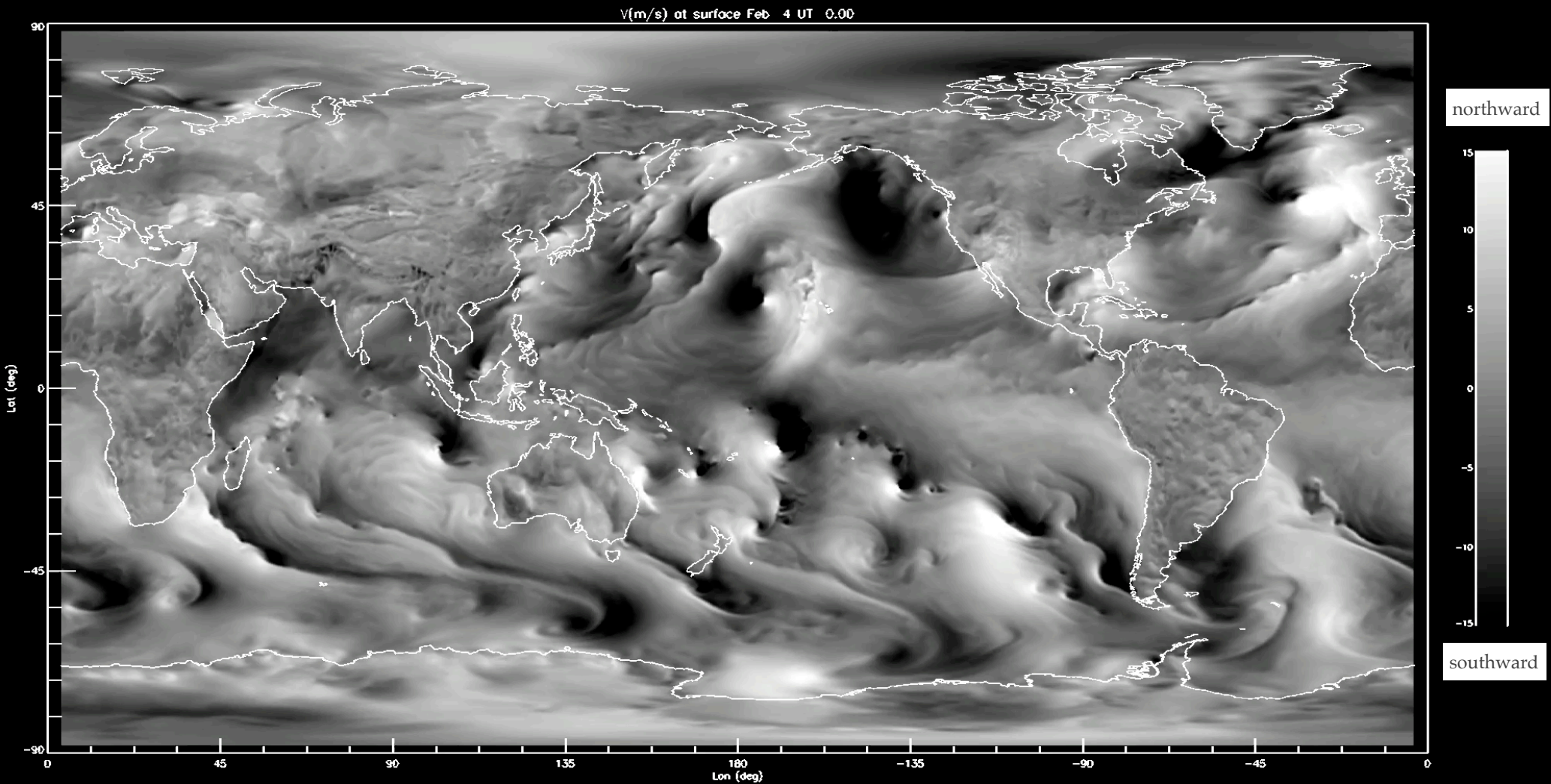
$$\frac{dw}{dt} + \frac{1}{\rho} \frac{\partial p}{\partial z} + g = 0,$$

$$\frac{1}{\rho} \frac{d\rho}{dt} + \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0,$$

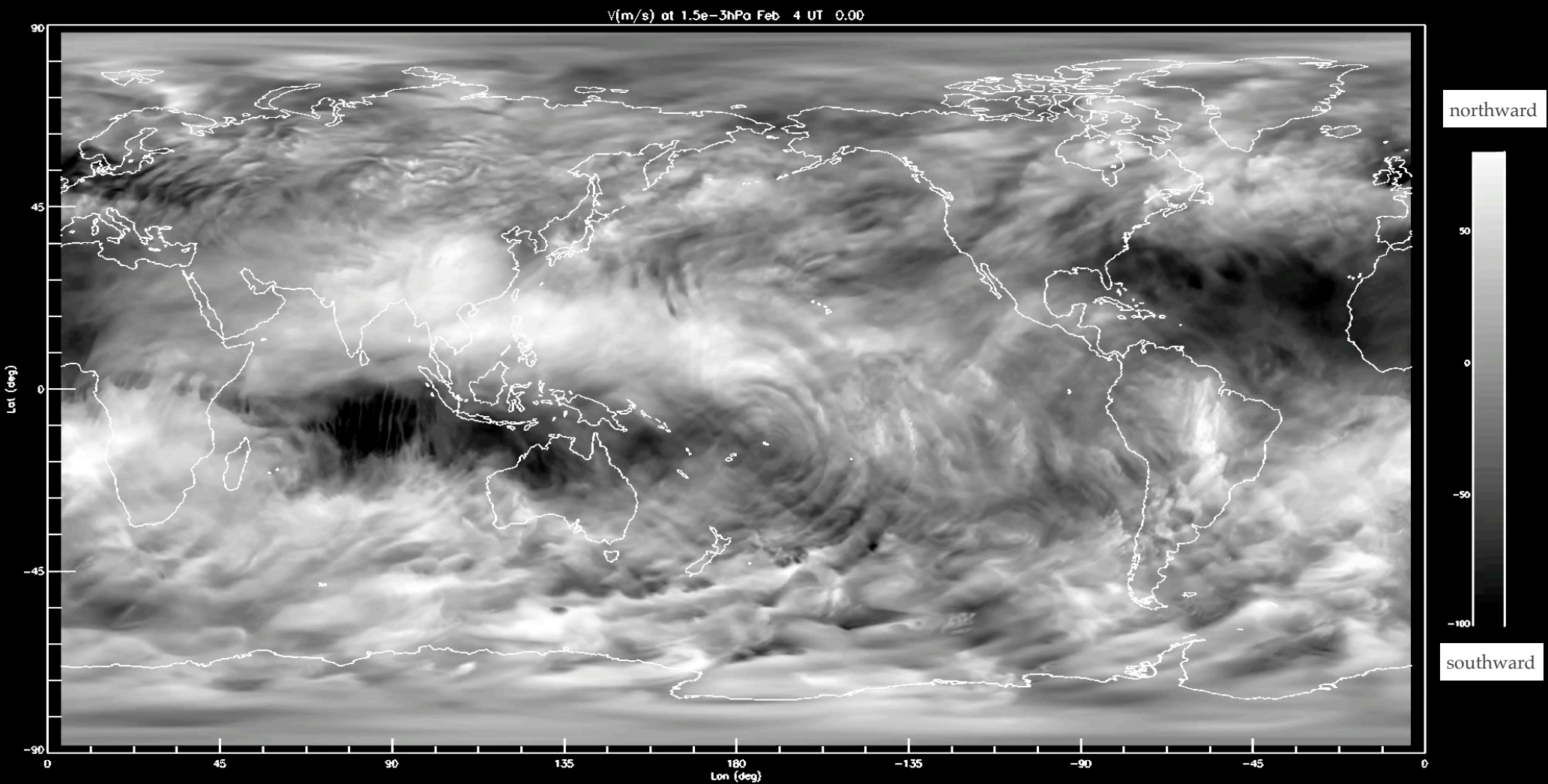
$$\frac{d\theta}{dt} = Q,$$



Meridional wind (m/s) from the same simulation, at Earth's surface (H. Liu et al., 2014)



Meridional wind (m/s) from the same simulation, at 100 km altitude (H. Liu et al., 2014)



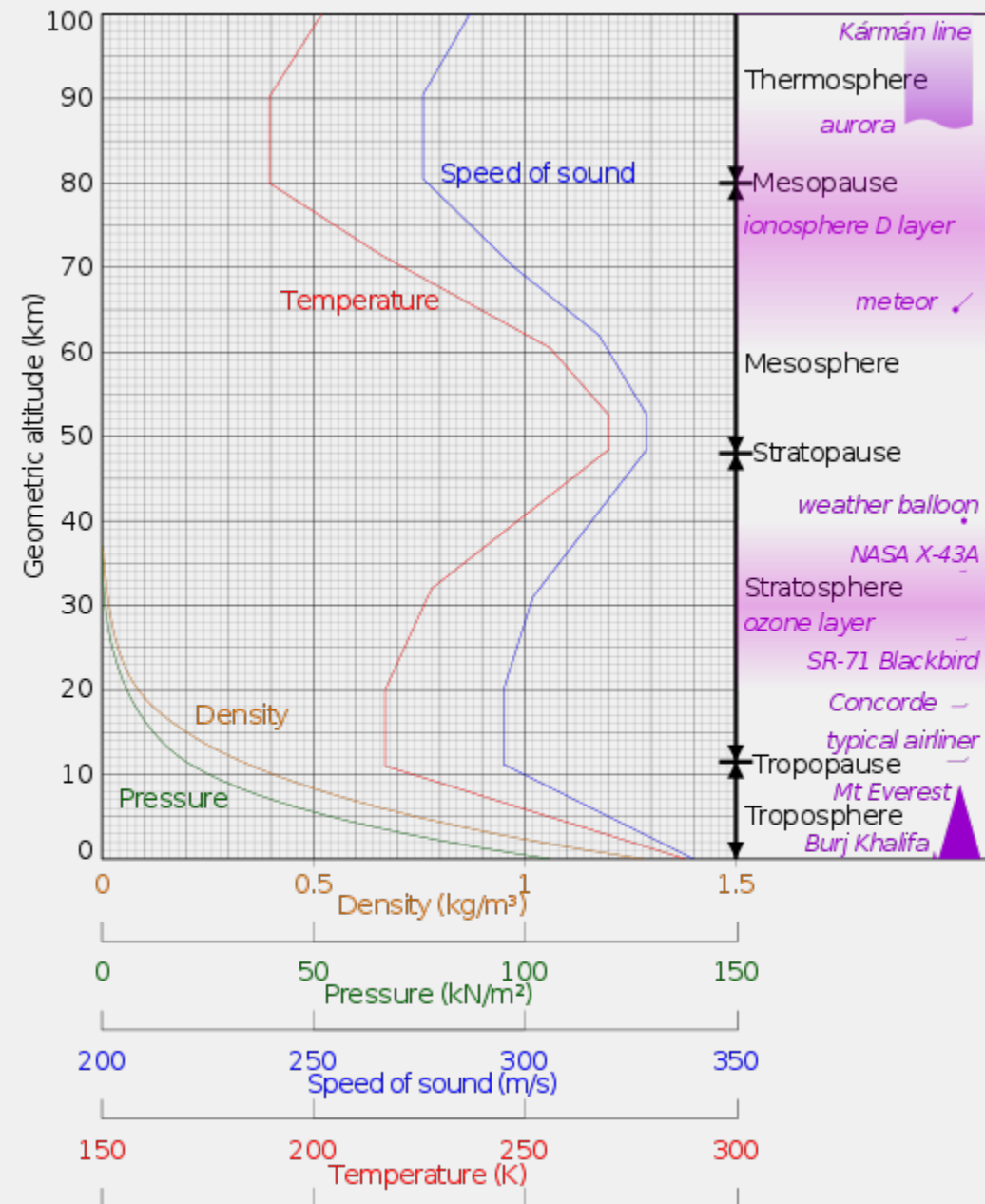
Why do all of these phenomena matter?

- ❖ **Science:** I just want to know how the atmosphere works.
Why? I just do.
- ❖ **Space Weather:** temperature changes, winds, waves, and tides all affect density and/or velocity of atmosphere, which affect spacecraft drag (as we'll see shortly)
 - ❖ as well as other effects on spacecraft and radio communications
- ❖ Winds / tides can also affect O/N₂ ratio, affecting drag and oxygen effects
- ❖ Gravity waves **break** at the bottom of the thermosphere; thought to cause **turbulence** that affects the height of the **turbopause**. But, as yet unverified!

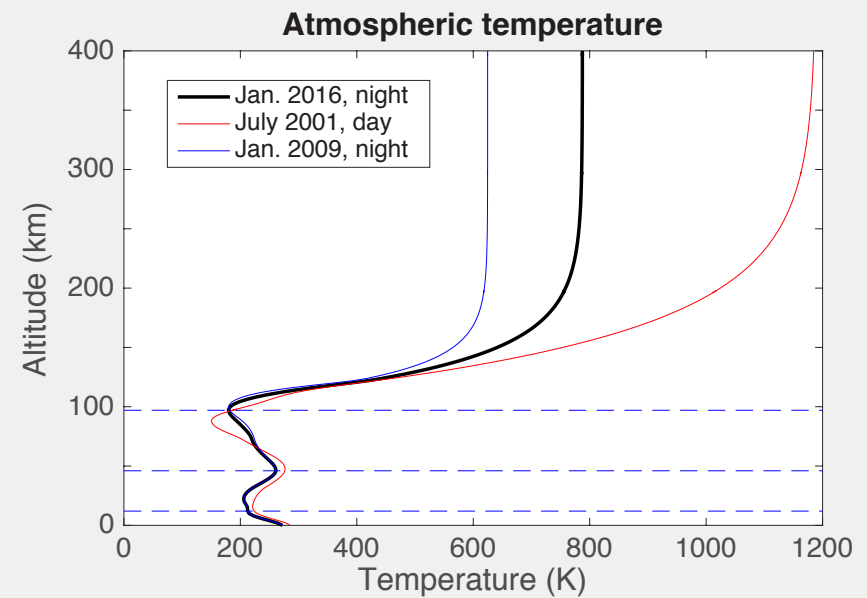
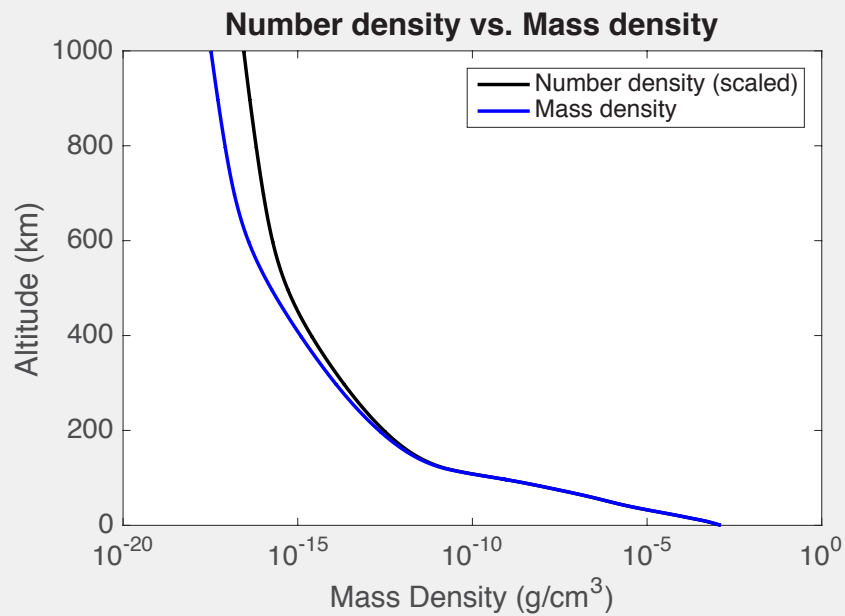
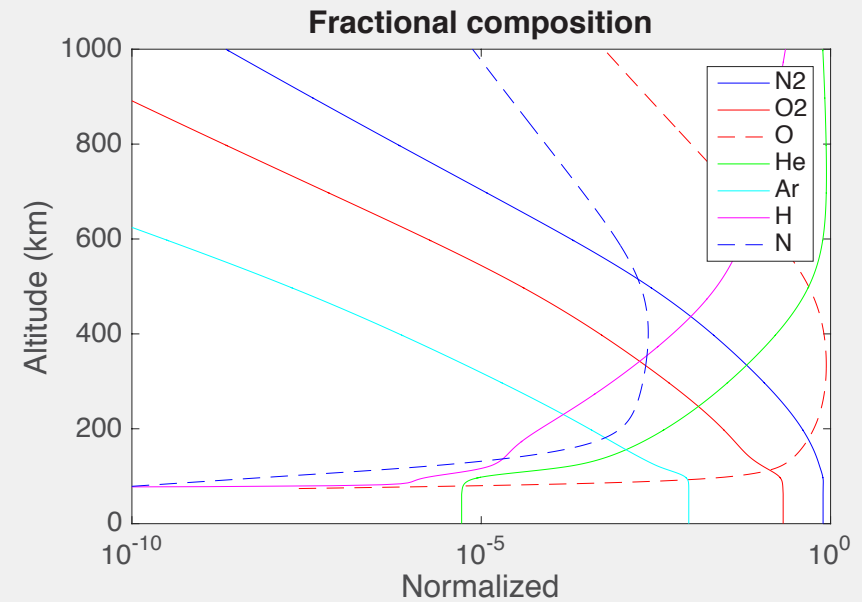
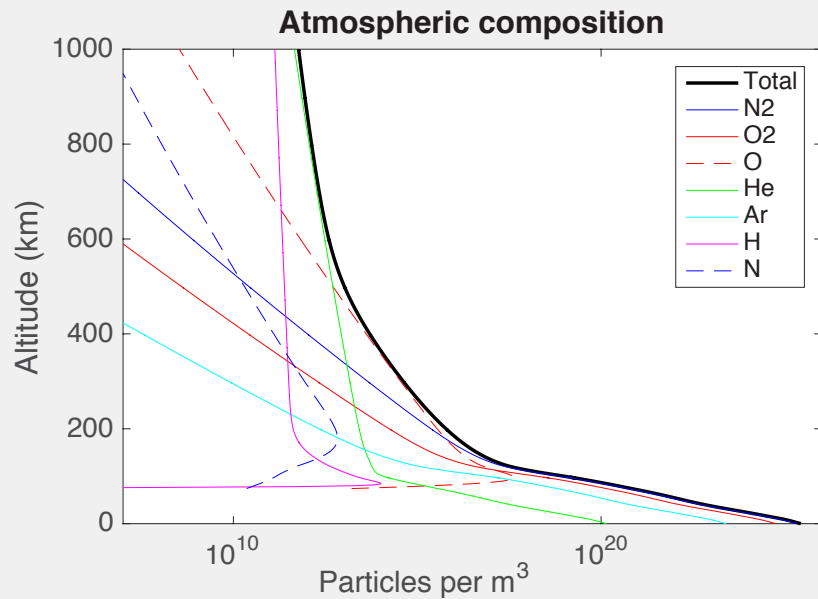
Atmospheric Models

- ❖ MSIS: Mass Spectrometer and Incoherent Scatter radar (horrible name)
- ❖ MSIS-E-90: <https://www.ukssdc.ac.uk/wdcc1/msise90.html>
- ❖ NRLMSISE00: <https://ccmc.gsfc.nasa.gov/modelweb/models/nrlmsise00.php>
- ❖ Global, empirical models of atmosphere density, composition, and temperature
- ❖ Derived from data: Mass spec, radar, rocket measurements, and more
- ❖ Include variations with solar activity, location, and more
- ❖ Other models: US Standard Atmosphere, International Standard Atmosphere, Jacchia, others

US Standard Atmosphere



Example output





University of Colorado **Boulder**

Atmospheric Effects

Currently Active Spacecraft

Satellite Quick Facts *(includes launches through 4/30/18)*

Total number of operating satellites: 1,886

United States: 859	Russia: 146	China: 250	Other: 631
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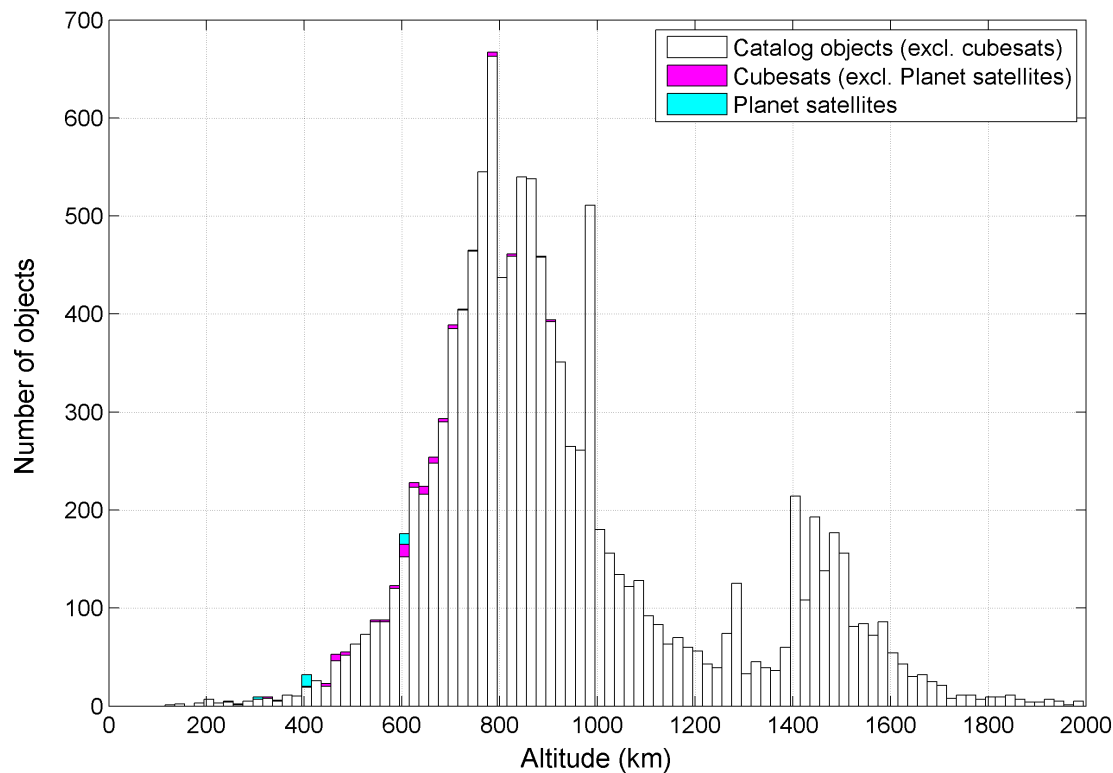
LEO: 1,186	MEO: 112	Elliptical: 40	GEO: 548
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Total number of US satellites: 859

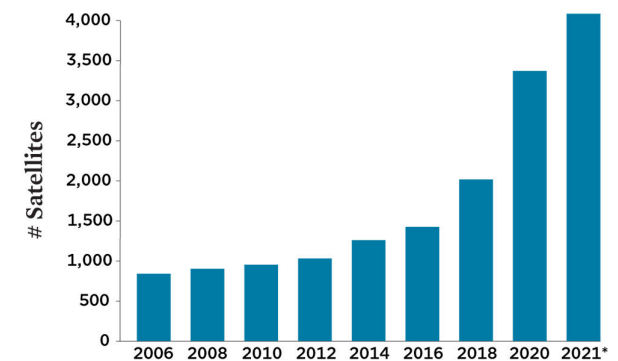
Civil: 20	Commercial: 495	Government: 178	Military: 166
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**Most spacecraft are
in LEO – subject to
atmosphere and drag**

Distribution of all objects including debris



Increase in Operating Satellites



* 2021 data covers January–April

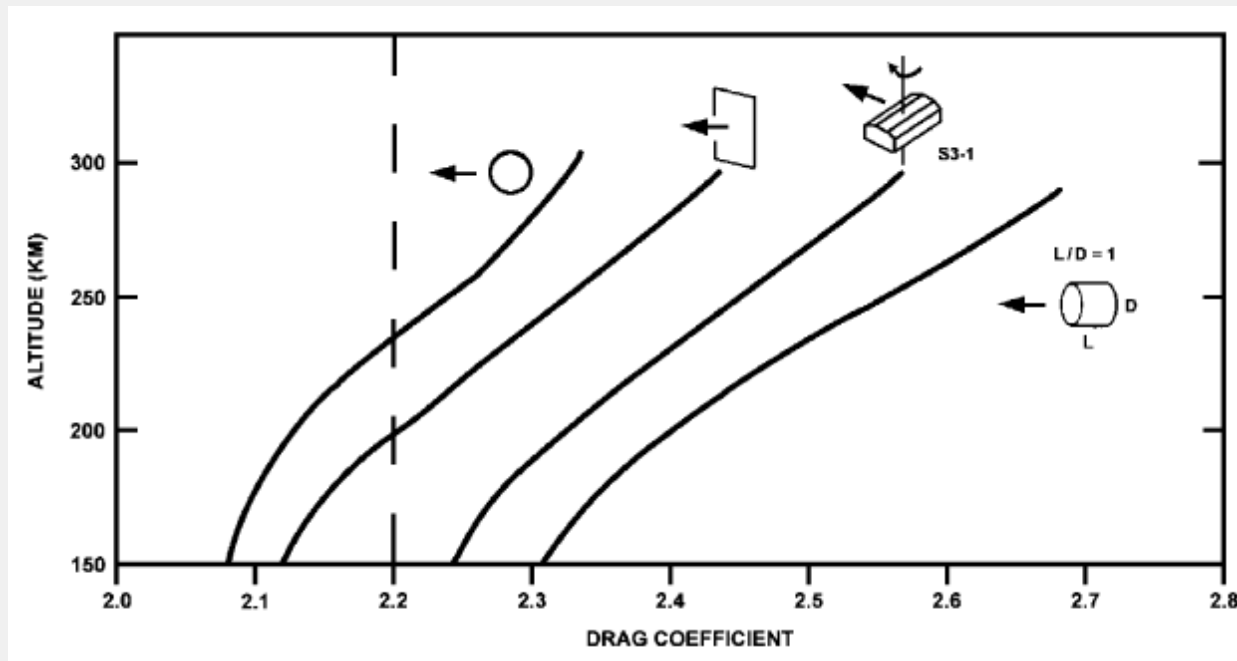
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Satellite Drag

- ❖ The atmosphere imposes a drag force on any object (just as you'd expect)

Drag coefficient

- ❖ Drag coefficient is particularly hard to measure!
- ❖ Lower-Left: drag coefficients in continuous flow regime
- ❖ In space: rarefied (free molecular) flow!

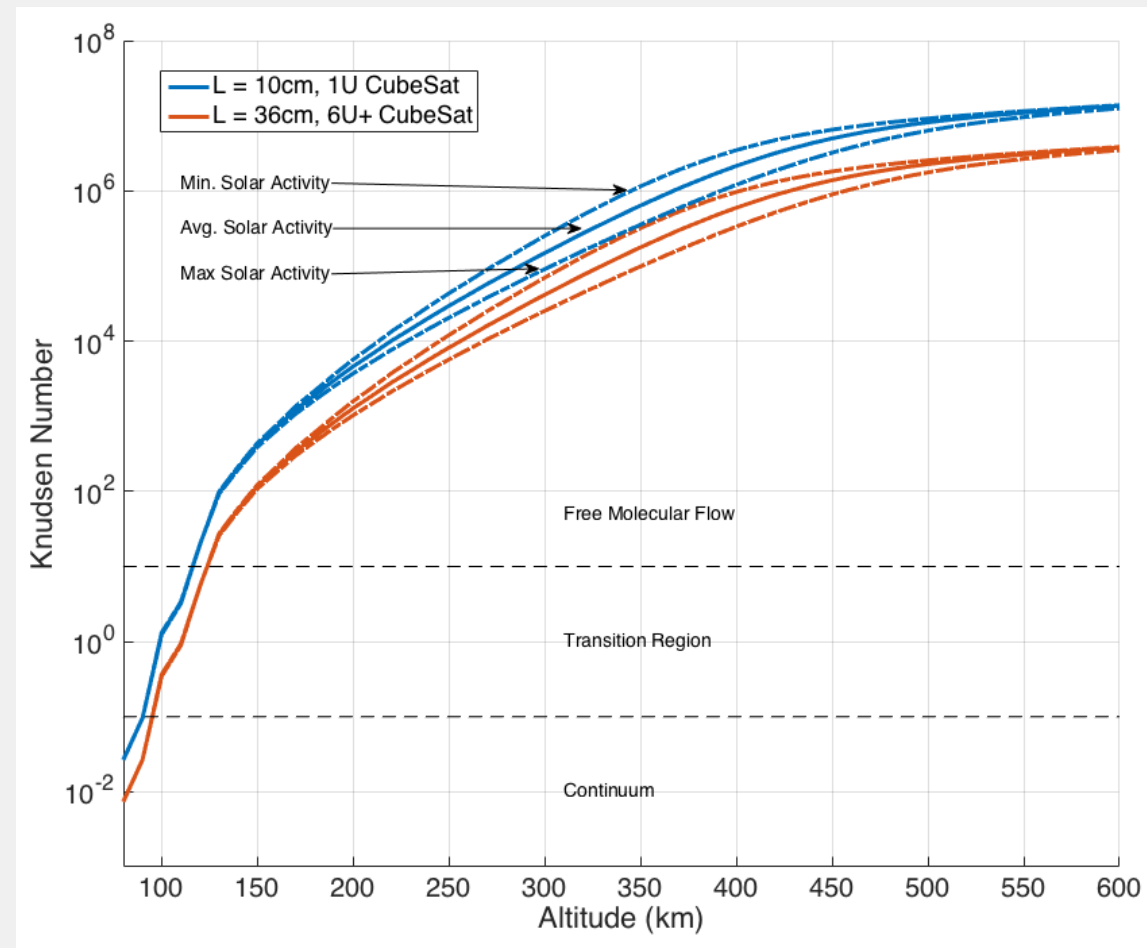
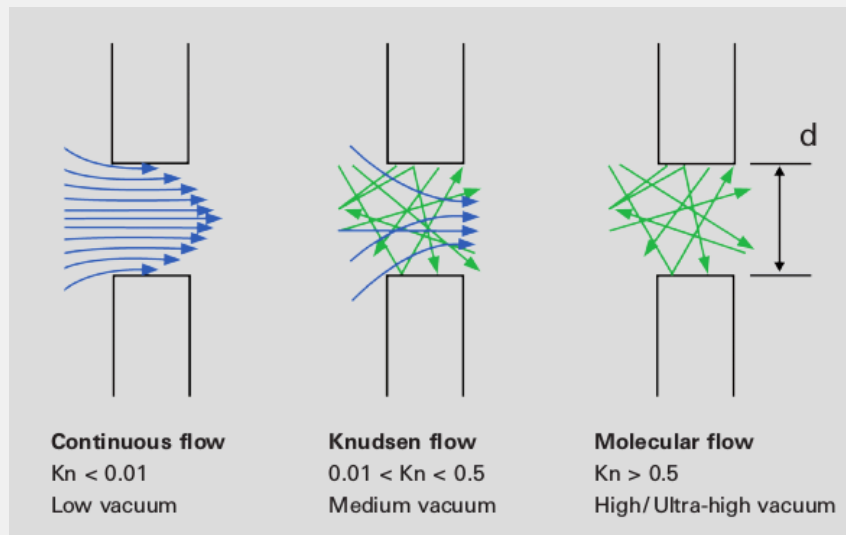


Shape	Drag Coefficient
Sphere → 	0.47
Halfsphere → 	0.42
Cone → 	0.50
Cube → 	1.05
Angled Cube → 	0.80
Long Cylinder → 	0.82
Short Cylinder → 	1.15
Streamlined Body → 	0.04
Streamlined Halfbody → 	0.09

Measured Drag Coefficients

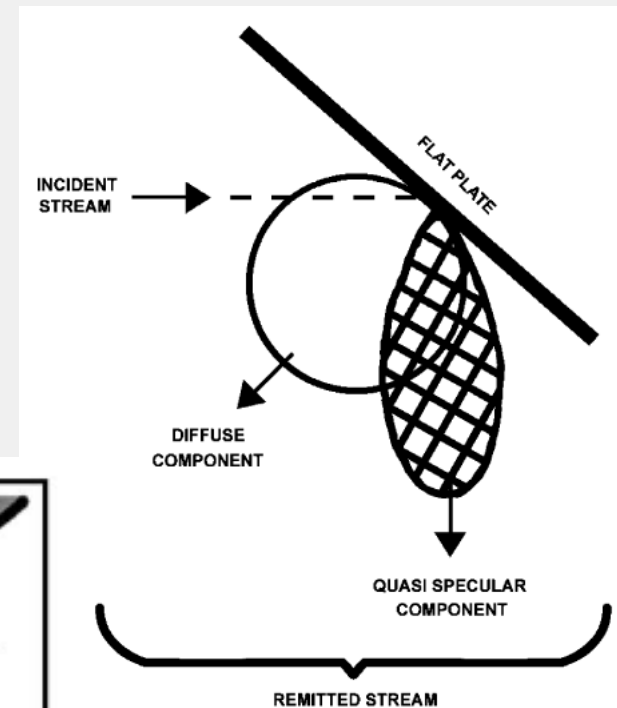
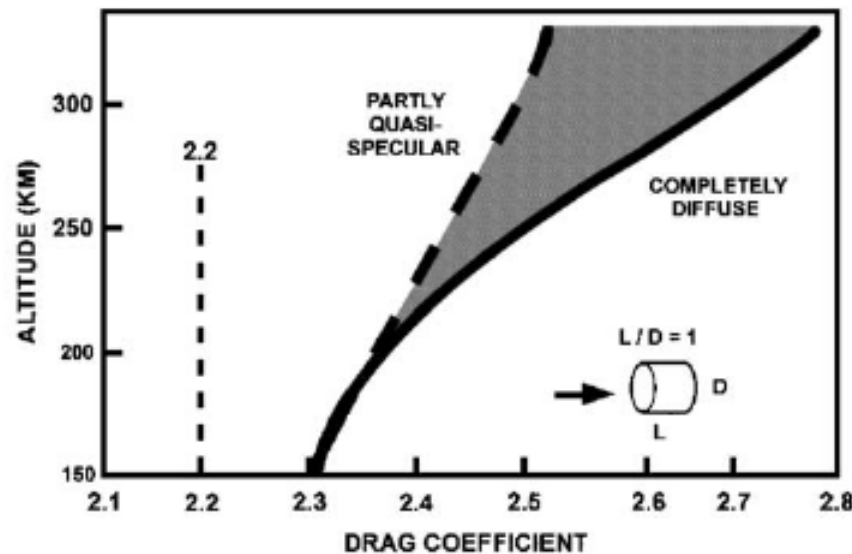
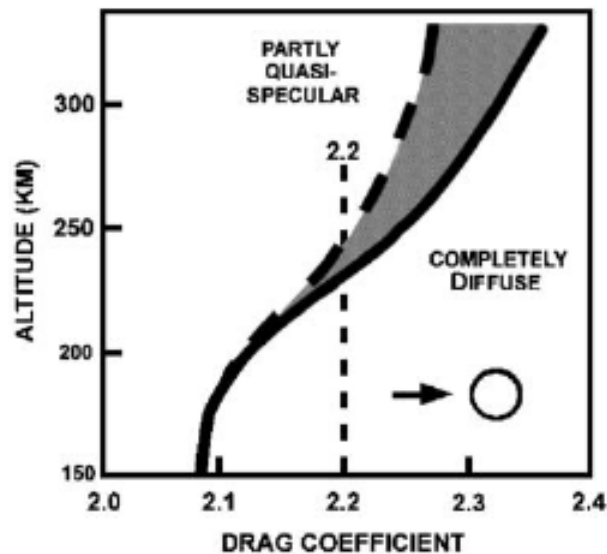
Continuous vs. Free Molecular Flow

- ❖ Knudsen number: $Kn = \lambda/L$
- ❖ $\lambda =$ **mean free path** (m)
between collisions
- ❖ $L =$ length scale (object)
- ❖ Basically, are we in a “flow”,
or just getting hit by
molecules / atoms one at a
time?



Drag coefficient and Scattering

- ❖ Drag coefficient depends on how molecules scatter from surfaces: either **specular** (mirror reflection) or **diffuse** (scattering from an opaque surface)
- ❖ This scattering is related to **accommodation**, which measures scattering elasticity
 - ❖ accommodation coefficient (≤ 1):



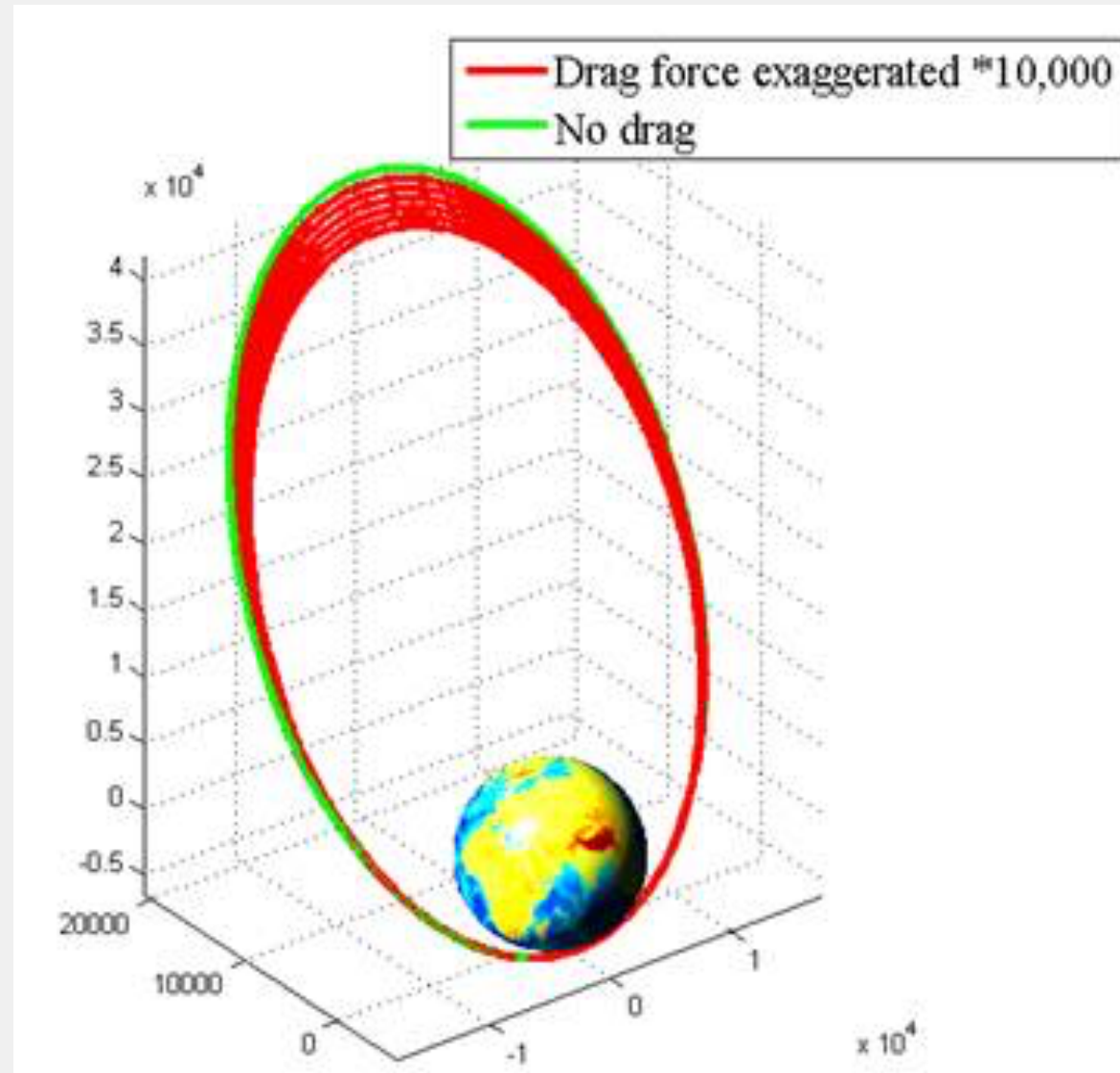
Does Drag make satellite speed up, or slow down?

- ❖ Kepler's third law relates period to altitude
(for circular orbit, semi-major axis $a = r$)

$$P^2 = \frac{4\pi^2}{GM} r^3 \quad v = \sqrt{\frac{GM}{r}}$$


Drag effect on elliptical orbit

- ❖ Drag force slows spacecraft down; now it can't reach as high of an apogee
- ❖ Returns to same perigee
- ❖ After a long time, orbit will be circular, with altitude = original perigee

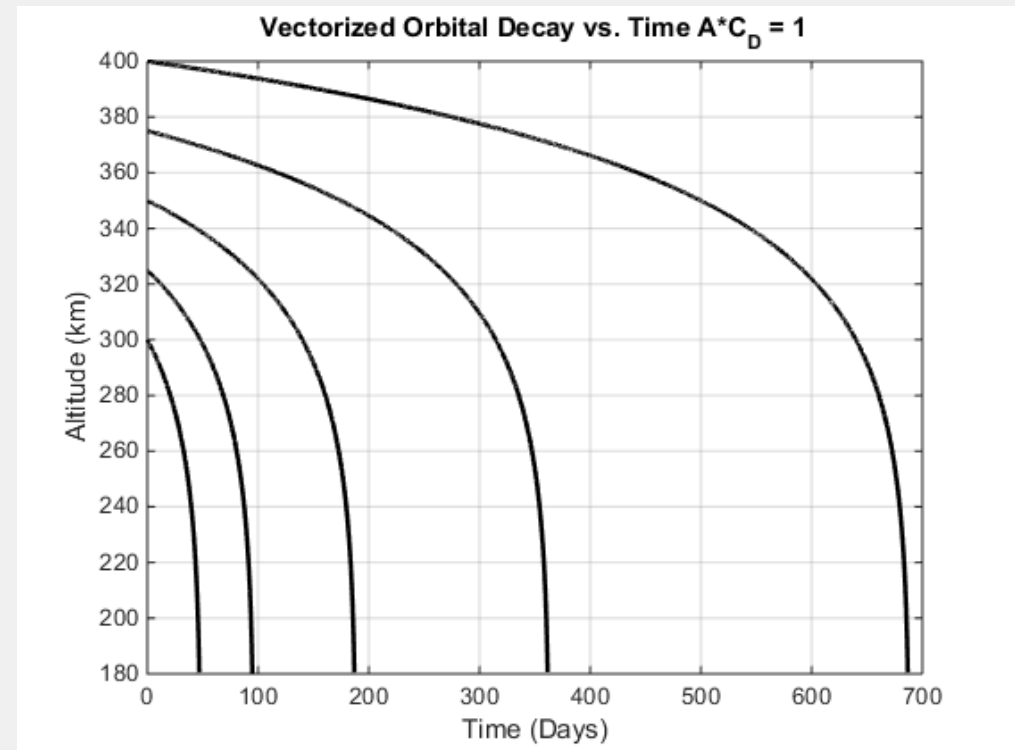


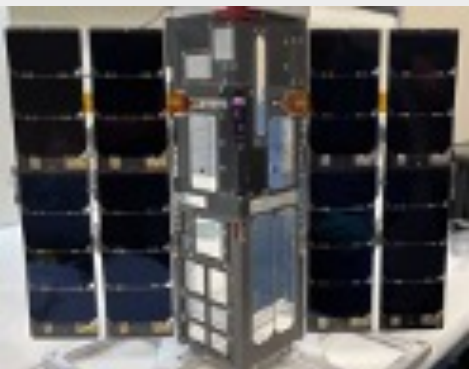
Magnitude of Drag Force

Orbit Degradation

- ❖ Differentiate Kepler's third law to determine the change in period with time (HW)
- ❖ period decreases with higher drag, higher area, density, and altitude
 - ❖ but density decreases with altitude
- ❖ Spacecraft mass (m) reduces change in period; "ballistic"
- ❖ From dP/dt , can also determine change in altitude; how long until it reaches some minimum altitude (e.g., 120km)?

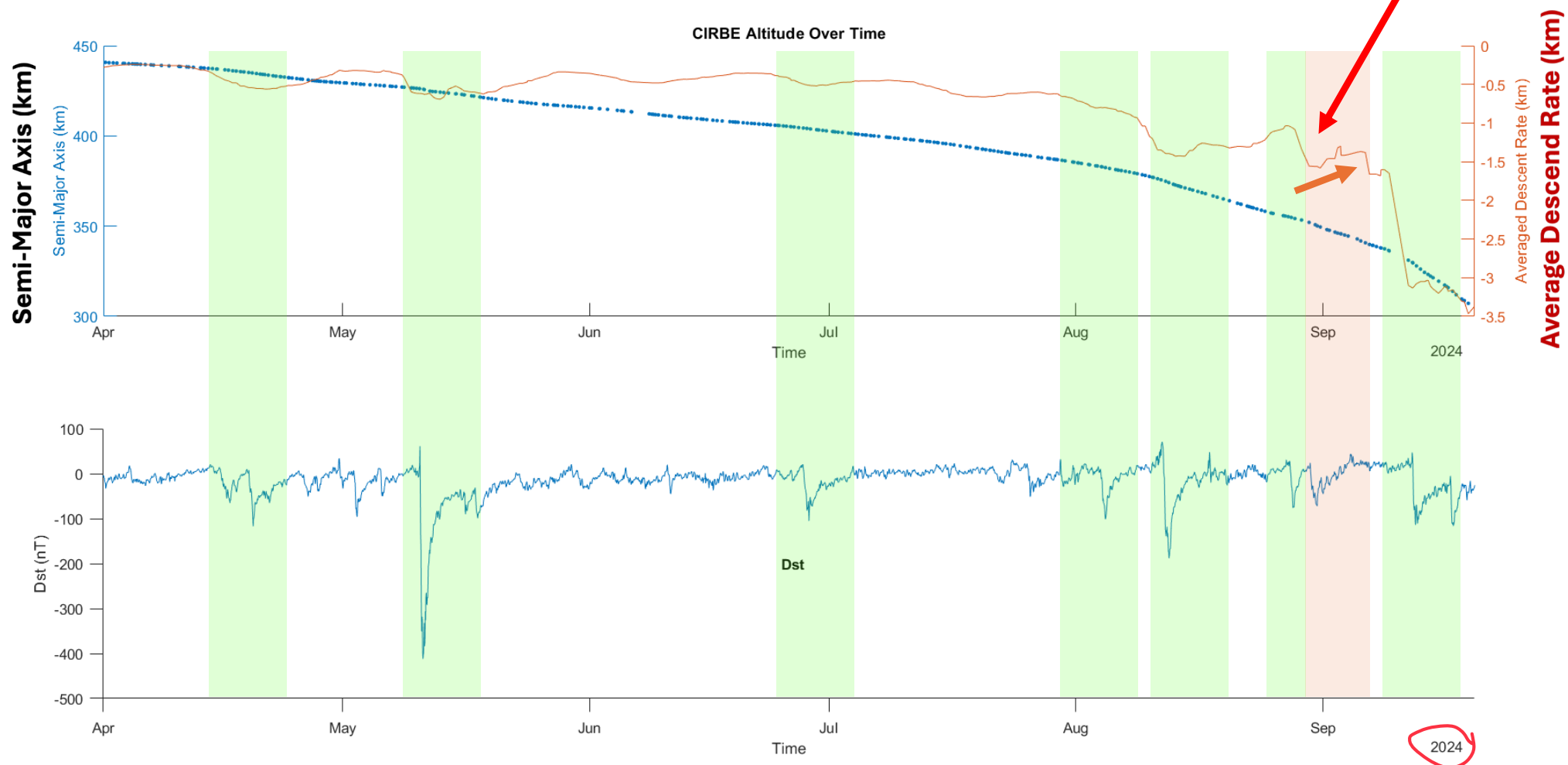
$$\frac{dP}{dt} = -3\pi r \rho \frac{C_D A}{m}$$





CIRBE's orbit altitude since April of 2024

Low Air-Drag
Flying Mode



ISS altitude vs time
